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Empirical energy levels and line lists of $D_2{}^{17}O$, and $D_2{}^{18}O$ from the analysis of far infrared absorption spectra (50–720 cm^{-1})

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ABSTRACT

This contribution is part of a long-term project aiming at improving our knowledge of the energy levels of the nine stable isotopologues of water vapor ($H_2{}^XO$, HD^XO, $D_2{}^XO$ with $X = 16$, 17, and 18) from the analysis of their far infrared (FIR) spectra. The Fourier transform spectra were recorded with unprecedented sensitivity at the SOLEIL synchrotron facility. In a recent work (S. N. Mikhailenko, A. O. Koroleva, A. Campargue *J. Phys. Chem. Ref. Data* 54, 023,103 (2025) https://doi.org/10.1063/5.0273106), the analysis of five absorption spectra of water vapor highly enriched in ^{17}O, ^{18}O and deuterium (D) have allowed a significant extension of the observations for the $HD^{17}O$ and $HD^{18}O$ minor isotopologues. It turns out that the $D_2{}^{17}O$ and $D_2{}^{18}O$ partial pressures in these spectra (up to about 0.38 mbar and 0.27 mbar, respectively) were about one order of magnitude higher than those in the SOLEIL spectra dedicated to the D_2O species (S. N. Mikhailenko, E. V. Karlovets, A. O. Koroleva, and A. Campargue *J. Phys. Chem. Ref. Data* 53, 023,102 (2024) https://doi.org/10.1063/5.0202355). As a result, 548 and 380 transitions are newly measured for $D_2{}^{17}O$ and $D_2{}^{18}O$, respectively, leading to the new determination of 147 and 59 energy levels, respectively. For each of the two species, the set of measured transition frequencies was merged to literature sources to generate extensive sets of accurate empirical rotational-vibrational energies for the ground, (000), and first excited, (010), vibrational states. Based on these accurate energy levels, recommended lists of transitions are generated with a 10^{-26} cm/molecule intensity cut-off for the (000) – (000), and (010) – (010) rotational bands and the ν_2 fundamental band of $D_2{}^{17}O$ and $D_2{}^{18}O$. For the microwave region (0 – 1.5 THz), more complete lists with an intensity cut-off lowered to 10^{-29} cm/molecule are provided.

1. Introduction

The $D_2{}^{18}O$ and $D_2{}^{17}O$ isotopologues of water vapor are the least abundant among the nine stable isotopologues of water vapor ("natural" relative abundance of 4.85×10^{-11} and 9.02×10^{-12}, respectively). As a result, the literature review indicates that the knowledge of their absorption spectra is very scarce (see Refs. [1,2]) and these isotopologues are the only ones missing in the list of water vapor transitions provided by the HITRAN database [3]. However, both species are included in the GEISA database (about 40,000 transitions between 1 and 12,740 cm^{-1}) [4]. A considerable extension of the $D_2{}^{18}O$ and $D_2{}^{17}O$ observations has been recently achieved in the far infrared (50 – 720 cm^{-1}) using four high quality spectra recorded by Fourier transform spectroscopy (FTS) at the AILES beam line of the SOLEIL synchrotron near Paris (https://www.synchrotron-soleil.fr/en) [2]. Among a series of twenty-one SOLEIL spectra of water vapor recorded with various isotopic compositions (#1-21), the four spectra analyzed in Ref. [2] (#15-18) were recorded with a highly enriched D_2O sample. Due to an additional partial enrichment in ^{17}O and ^{18}O, $D_2{}^{17}O$ and $D_2{}^{18}O$ partial pressures reach values of 22.4 μbar and 33.6 μbar, respectively. This allowed us for measuring for the first time 1089 $D_2{}^{17}O$ transitions in the 50-720 cm^{-1} region [2]. For $D_2{}^{18}O$, more than 1150 transitions were measured while only 128 transitions (up to 216.37 cm^{-1}) were previously known from Ref. [5] in the region.

The series of spectra recorded at SOLEIL synchrotron includes five spectra (#10-14) corresponding to a high enrichment in both ^{17}O, ^{18}O oxygen and deuterium. Their isotopic composition was optimized for the study of the $HD^{18}O$ and $HD^{17}O$ species, recently reported in Ref. [6]. Nevertheless, it turned out that the $D_2{}^{17}O$ and $D_2{}^{18}O$ partial pressures in these spectra (up to about 0.38 mbar and 0.27 mbar in spectrum #14,

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respectively) were about one order of magnitude larger than their maximum values in spectra #15-18 of Ref. [2], allowing for additional observations for these species. The present contribution is thus devoted to the analysis of the $D_2{}^{17}O$ and $D_2{}^{18}O$ transitions observed in the spectra #10-14. Our general aim is to present the sets of new $D_2{}^{17}O$ and $D_2{}^{18}O$ observations completing those of Ref. [2], check the consistency with the results of Ref. [2], merge the SOLEIL observations with literature sources to generate improved sets of accurate empirical rotational-vibrational energies for the ground and first excited vibrational states, and generate recommended lists of transitions for the (000) – (000), and (010) – (010) rotational bands and the ν_2 fundamental band of $D_2{}^{17}O$ and $D_2{}^{18}O$.

2. Experiment and line list

The reader is referred to Ref. [6] for details on the recording procedure of spectra #10-14 and on the construction of the corresponding experimental lists. Briefly, the recordings covering the 50 – 720 cm^{-1} range were obtained at room temperature (295.5±0.3 K) using a Bruker 125 interferometer equipped with a multipass absorption cell in a White-type configuration providing a total absorption path length of 151.75±1.5 m. The spectra #10-12 at low pressure were recorded with a maximum spectral resolution of 0.001 cm^{-1} while spectra #13 and #14 at 1.95 and 8.03 mbar, respectively, were recorded with a spectral resolution of 0.002 cm^{-1}. The isotopically enriched water sample was obtained by mixing a similar amount of a D_2O sample (from Sigma-Aldrich, deuterium enrichment larger than 99.96%) with a sample of water highly enriched in ^{17}O (from Eurisotop). According to the certificate of analysis, the latter has oxygen isotopic abundances of 57.9, 38.2 and 1.2 % for ^{17}O, ^{16}O and ^{18}O, respectively. As a result of the statistical distribution of oxygen and hydrogen atoms among the different water molecules, the nine stable water isotopologues ($H_2{}^XO$, HD^XO, $D_2{}^XO$ with X = 16, 17, and 18) are present in the mixture with a statistical relative abundance. The isotopic exchanges with water molecules adsorbed in the walls of the cell leads to distinct isotopic compositions for the five analyzed spectra. (The isotopic abundances were derived by direct comparison to simulations using SPECTRA system (https://spectra.iao.ru/) and HITRAN's water line lists [3] for the first seven isotopologues and Schwenke-Partridge calculated line lists (SP) for $D_2{}^{17}O$ and $D_2{}^{18}O$ [7,8]). The obtained abundance values of the $D_2{}^XO$ isotopologues are listed in Table 1.

A global experimental line list gathering the observations of spectra #10-14 was elaborated by combining the line list of the five individual spectra (see details in Ref. [6]). The individual line lists were elaborated from a multiline fitting of the transmittance spectra #10-14 assuming the standard Voigt line profile as the line shape (no particular care was taken for the treatment of the apparatus function). Depending on the line intensity for each line, the parameters retrieved from the spectrum corresponding to the best experimental conditions was selected for the final global line list (see Fig. 2 in Ref. [6]). The resulting global line list counting a total of more than 13100 entries has been provided as supplementary material of Ref. [6]. Here, we provided as supplementary material (**SM1**) the line parameters limited to the three $D_2{}^XO$ isotopologues. The selected source (#10-14) of the line parameters is indicated for each line together with estimated values of the position and intensity uncertainties as provided by the fit. For a significant fraction of the lines, the fit uncertainty on the position was found to be smaller than 5×10^{-5} cm^{-1}. Such small values which are believed to underestimate the real uncertainty on the line position, were replaced by a default value of 5×10^{-5} cm^{-1}. This minimum uncertainty value is consistent with the estimated accuracy of our frequency calibration (the absolute frequency calibration was performed using about 300 $H_2{}^{16}O$ lines (ν^{obs}) with positions (ν^{ref}) known with an accuracy better than 1×10^{-6} cm^{-1} [9]. After a linear correction of the frequency scale, an *RMS* deviation of 2.85×10^{-5} cm^{-1} was obtained for the (ν^{obs} - ν^{ref}) position differences. Note that the large range in the pressures of the recordings (about a factor 500 – see Table 1) allowed measuring lines with intensities spanning more than five orders of magnitude (from about 10^{-24} to more than 10^{-19} cm/molecule for pure species). As discussed in Ref. [6], the accuracy on the determined intensity value does not allow competing with the high accuracy of *ab initio* calculations (typically 1%). Thus, in the following, we will mostly focused on line positions.

The studied FIR region corresponds to the purely rotational transitions in the ground vibrational state and weaker rotational transitions in the excited vibrational states, namely (010), (100), (020), and (001), which are slightly populated at room temperature. A total of 2147, 1624 and 1523 transitions were assigned to the $D_2{}^{16}O$, $D_2{}^{17}O$ and $D_2{}^{18}O$ isotopologues, respectively. The position comparison of these observations to previous SOLEIL measurements of Refs. [2,10]) is discussed below and included in the D_2O line lists provided as supplementary material.

Table 1
Experimental conditions and D_2O relative abundances of the five FTS spectra recorded at SOLEIL under analysis. The temperature was 295.5 K.

Label	Pressure mbar	Resolution cm^{-1}	Number of scans	Relative abundance (%)		
				$D_2{}^{16}O$	$D_2{}^{18}O$	$D_2{}^{17}O$
#10	0.06 [a]	0.001	160	6.4	1.2	1.9
#11	0.29	0.001	160	15.4	2.1	3.6
#12	0.015 [a]	0.001	300	7.0	1.4	1.5
#13	1.97	0.002	160	26.4	3.0	5.1
#14	8.03	0.002	220	26.7	3.4	4.8

[a] Estimated value – see text (the used pressure gauge does not allow measuring so low pressures).

3. $D_2{}^{16}O$

The #15-18 SOLEIL spectra of highly enriched D_2O samples analyzed in Ref. [2] allowed for a considerable extension of the observations of the three D_2O isotopologues between 50 and 720 cm^{-1}. In the case of $D_2{}^{16}O$, the isotopic abundance was very high (up to 96.2%) and 2057 of the 2886 measured $D_2{}^{16}O$ positions were newly reported compared to literature. In the present work, the maximum $D_2{}^{16}O$ abundance is 26.7 % (see Table 1) allowing for only marginal additional information compared to Ref. [2]. Over a total of 2147 $D_2{}^{16}O$ transitions presently measured, only four lines are newly reported. For 2142 line positions in common between this work and Ref. [2] ($\nu^{\#15\text{-}18}$), an $RMS = \sqrt{\sum_{i=1}^{N}\left(\nu_i^{TW} - \nu_i^{\#15-18}\right)^2/N}$ value of 12.3×10^{-5} cm^{-1} is obtained with a maximum discrepancy $d\nu = |\nu^{TW} - \nu^{\#15\text{-}18}|$ of 119×10^{-5} cm^{-1}. Only 13 position deviations corresponding to very weak lines or unresolved doublets exceed 50×10^{-5} cm^{-1}. At least three of these positions are more precise in the present spectra #10-14. Five line positions presently measured could not be retrieved from the spectra #15-18 analyzed in Ref. [2] due to their overlapping with stronger lines. Note that these positions do not involve new energy levels and show a very good agreement with their empirically predicted values included in the recommended line list of Ref. [2] (maximum discrepancy of 53×10^{-5} cm^{-1} for the weakest transition (010) 16 $_{7\ 9}$ – (010) 16 $_{4\ 12}$ at 261.70377(47) cm^{-1}).

A large set of 2152 $D_2{}^{16}O$ transitions was also obtained during the analysis of spectra #19-21 dedicated to the $HD^{16}O$ species [10]. The corresponding $RMS = \sqrt{\sum_{i=1}^{N}\left(\nu_i^{TW} - \nu_i^{\#19-21}\right)^2/N}$ value is 14.3×10^{-5} cm^{-1} for $N = 2022$ transitions in common. The maximum discrepancy $d\nu = |\nu^{TW} - \nu^{\#19\text{-}21}|$ is 141×10^{-5} cm^{-1} for the 13 $_{11\ 2}$ – 13 $_{8\ 5}$ transition at 461.18926 cm^{-1}. The 31 position deviations exceeding 50×10^{-5} cm^{-1} correspond either to weak or overlapped lines or unresolved doublets.

4. $D_2{}^{17}O$

4.1. Line positions

The overview of the assigned $D_2{}^{17}O$ transitions is presented in Fig. 1 and compared to the $D_2{}^{17}O$ list of Ref. [2]. As the experimental intensity values are strongly underestimated in the case of the strongest lines, the SP variational intensities (for pure $D_2{}^{17}O$) have been used for the plot. Overall the intensities of the observed transitions span over six orders of magnitude. The maximum $D_2{}^{17}O$ partial pressure in spectra #15-18 of Ref. [2] was about 22.4 μbar (corresponding to a 0.28% relative abundance at 8 mbar in spectrum #17). In the presently analyzed spectrum #14, the $D_2{}^{17}O$ partial pressure reaches a value of about 384 μbar (4.8% at 8 mbar – see Table 1). As a result, the set of 1088 $D_2{}^{17}O$ transitions of Ref. [2] is presently increased to 1624 transitions, 548 transitions being new compared to all the previous SOLEIL analysis (538 $D_2{}^{17}O$ transitions were also measured in Ref. [10]). For the ν_2-ν_2 rotational band, the new observations add 150 transitions to the set of 224 transitions reported in Ref. [2] . Seven transitions of the $2\nu_2$-$2\nu_2$ rotational band are reported. These are the first observations of transitions of this band. Due to accidental overlapping with stronger lines, twelve $D_2{}^{17}O$ transitions reported in Ref. [2] are not observed in the present spectra.

The comparisons of the line positions observed in this study to those of Ref. [2] ($\nu^{\#15\text{-}18}$) and Ref. [10] ($\nu^{\#19\text{-}21}$) give a very satisfactory agreement (see supplementary material **SM1**). We obtain an $RMS = \sqrt{\sum_{i=1}^{N}\left(\nu_i^{TW} - \nu_i^{\#15-18}\right)^2/N}$ value of 12.3×10^{-5} cm^{-1} for $N = 1074$ transitions in common with a maximum discrepancy $d\nu = |\nu^{TW} - \nu^{\#15\text{-}18}|$ of 101×10^{-5} cm^{-1} for the $5_{0\,5} - 4_{1\,4}$ transition at 51.62985 cm^{-1}. Discrepancies exceed 50×10^{-5} cm^{-1} for only eleven weak and noisy lines or unresolved doublets. The two largest disagreements of -0.00997 and -0.12009 cm^{-1} (not included in the above statistics) correspond to the $21_{3\,18} - 20_{4\,17}$ and $18_{7\,12} - 17_{6\,11}$ pure rotational transitions, respectively and are due to assignment errors: *(i)* in Ref. [2], the line at 307.97083 cm^{-1} was erroneously assigned to the $18_{7\,12} - 17_{6\,11}$ pure rotational transition while it is due to the $HD^{16}O$ pure rotational doublet $12_{12\,1} - 12_{11\,2}$ and $12_{12\,0} - 12_{11\,1}$ [10]. The correct position of the $D_2{}^{17}O$ $18_{7\,12} - 17_{6\,11}$ transition corresponds to the line at 307.85065 cm^{-1} which was left unassigned in spectrum #15 of Ref. [2]. Note that this is the only transition of spectra #15-18 involving the (000) $18_{7\,12}$ energy level. In this work, the assignment of the $18_{7\,12} - 17_{6\,11}$ transition to the line at 307.97083 cm^{-1} is confirmed by the involvement of the $18_{7\,12}$ level in four additional transitions: as upper level of the $18_{7\,12} - 18_{4\,15}$ transition at 342.69757 cm^{-1} and of the $18_{7\,12} - 17_{4\,13}$ transition at 446.57337 cm^{-1} and as lower level of the $18_{8\,11} - 18_{7\,12}$ and $19_{8\,11} - 18_{7\,12}$ transitions at 121.85972 cm^{-1} and 356.44352 cm^{-1}, respectively; *(ii)* the line at 233.11407 cm^{-1} assigned to the $21_{3\,18} - 20_{4\,17}$ pure rotational transition of $D_2{}^{17}O$ in Ref. [2] corresponds in fact to the $15_{6\,9} - 14_{6\,8}$ pure rotational transition of $HD^{17}O$. The correct position of the $D_2{}^{17}O$ $21_{3\,18} - 20_{4\,17}$ transition presently determined from spectrum #13 is 233.10410 cm^{-1}. This assignment is confirmed by two additional transitions involving the $21_{3\,18}$ upper level: $21_{3\,18} - 21_{2\,19}$ and $21_{3\,18} - 20_{2\,19}$ at 164.68777 and 561.17804 cm^{-1}, respectively.

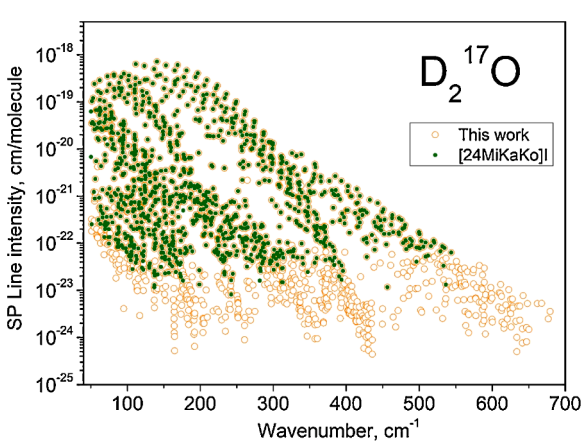

Fig. 1. Overview of the $D_2{}^{17}O$ transitions measured in this work (spectra #10–14, orange circles) and in Ref [2] ([24MiKaKo]I) (spectra #15–18, green dots). The measured line positions have been associated with the SP intensities (for pure $D_2{}^{17}O$).

For completeness, the positions comparisons provided as a supplementary material (**SM1**) include the comparison to the 537 $D_2{}^{17}O$ positions measured in Ref. [10]. Again, the overall agreement is satisfactory with an $RMS = \sqrt{\sum_{i=1}^{N}\left(\nu_i^{TW} - \nu_i^{\#19-21}\right)^2/N}$ value of 14.7×10^{-5} cm^{-1} and a maximum discrepancy $d\nu = |\nu^{TW} - \nu^{\#19\text{-}21}|$ of 98×10^{-5} cm^{-1} (for the $6_{6\,0} - 5_{3\,3}$ transition at 310.22917 cm^{-1}).

4.2. Energy levels

The significant extension of the set of FIR line positions of $D_2{}^{17}O$ allows for an update of the empirical energy levels reported in Ref. [2]. As detailed in this reference, only four pure rotational transitions from Ref. [11] were known in the region prior to the analysis of the SOLEIL spectra. A total of 724 empirical energy levels were determined by merging the 1088 position measurements of Ref. [2] with previous observations in other spectral regions: 219 microwave transitions given by Bellet et al. [12] and Puzzarini et al. [11,13], 226 ν_2 transitions by Toth [14] and 313 transitions of five bands in near infrared [15,16]. The RITZ computer code developed by S.A. Tashkun (IAO, Tomsk) [17–20] was used for the energy determination. Note that seven ν_2-ν_2 rotational transitions observed in Ref. [2] were not included in the main spectroscopic network, leaving undetermined the energies of the ten levels involved in these transitions.

The set of line positions of Ref. [2] can be merged with more than 2150 line positions using the recent SOLEIL results from spectra #10-14 (TW, this work) and spectra #19-21 [10]. This allowed us to derive a total of 871 term values of seven vibrational states, 147 being newly determined (see Table 2). Note that three pure rotational transitions, four ν_2-ν_2 transitions and the seven $2\nu_2$-$2\nu_2$ transitions are not included in the main spectroscopic network. So, the energies of the 26 levels involved in these transitions [(000): $24_{1\,23}$, $24_{2\,23}$, $24_{3\,22}$, $25_{1\,24}$, $25_{2\,24}$, and $25_{2\,23}$; (010): $13_{8\,5}$, $14_{9\,6}$, $17_{4\,14}$, $18_{3\,15}$, $20_{0\,20}$, $20_{1\,20}$, $21_{0\,21}$, and $21_{1\,21}$; (020): $5_{5\,1}$, $5_{5\,0}$, $6_{5\,2}$, $6_{5\,1}$, $6_{6\,1}$, $6_{6\,0}$, $7_{6\,2}$, $7_{6\,1}$, $8_{5\,3}$, $8_{7\,2}$, $8_{7\,1}$, and $9_{6\,4}$] could not be derived.

The set of $D_2{}^{17}O$ empirical energy levels of the ground and (010) states is provided as a separate supplementary material (**SM2**) which includes the comparison to the previous set of Ref. [2]. The present set contains 458 term values for the ground vibrational state up to $J_{max} = 27$, $K_{a\,max} = 17$ and $E_{max} = 4101$ cm^{-1} and 250 term values for the first excited vibrational state up to $J_{max} = 19$, $K_{a\,max} = 13$ and $E_{max} = 3788$ cm^{-1}. Compared to Ref. [2], 82 ground state energy levels are newly determined and two are corrected. For the first excited state, (010), 65 levels are newly determined and three are corrected. The comparison of the $D_2{}^{17}O$ empirical levels in common shows a very good agreement for the energies with an *RMS* of 8.4×10^{-5} cm^{-1} for the $dE = E^{TW} - E^{Ref.[2]}$ energy differences ($N = 556$). The exceptions (not included in the statistics) are the energies of the above discussed $18_{7\,12}$ and $21_{3\,18}$ levels of the ground state and of three levels ($12_{3\,9}$, $13_{4\,10}$ and $14_{3\,11}$) of the (010) state. The disagreement of about 0.06 cm^{-1} for these three (010) levels is due to an erroneous assignment of the line at 301.73183 cm^{-1} to the $D_2{}^{17}O$ (010) $12_{3\,9}$ – (010) $11_{2\,10}$ transition (as second assignment) in spectrum #15 of Ref. [2] while it corresponds to the $HD^{18}O$ line (010) $11_{4\,7}$ – (010) $10_{3\,8}$. The correct position of the $D_2{}^{17}O$ (010) $12_{3\,9}$ – (010) $11_{2\,10}$ transition is presently determined at 301.79137 cm^{-1}.

Table 2
Empirical energies of newly determined rotational levels of $D_2{}^{17}O$ in the ground and first excited vibrational states.

Vib	J	K_a	K_c	E^{emp}	dE
000	15	15	1	3232.17498	19
000	15	15	0	3232.17506	35
000	16	15	2	3432.13558	26
000	16	15	1	3432.13550	13
000	16	16	1	3637.35056	61
000	16	16	0	3637.35048	57
000	17	11	6	2890.49572	37
000	17	14	4	3443.39535	27
000	17	14	3	3443.39541	25
000	17	15	3	3643.73390	19
000	17	15	2	3643.73393	25
000	17	16	2	3850.15034	28
000	17	16	1	3850.15042	35
000	17	17	1	4061.51744	59
000	17	17	0	4061.51752	69
000	18	9	10	2794.91871	17
000	18	11	8	3110.66489	25
000	18	12	7	3286.17251	32
000	18	13	6	3471.53691	26
000	18	13	5	3471.53679	20
000	18	14	5	3665.49113	33
000	18	14	4	3665.49106	22
000	18	15	4	3866.84449	47
000	18	15	3	3866.84443	28
000	18	16	3	4074.45064	46
000	18	16	2	4074.45061	31
000	19	5	14	2631.38802	20
000	19	6	13	2702.71113	16
000	19	8	11	2892.50768	19
000	19	9	11	3026.88921	17
000	19	9	10	3027.08357	20
000	19	10	10	3178.05187	21
000	19	10	9	3178.06617	31
000	19	11	9	3342.40072	26
000	19	11	8	3342.40025	25
000	19	12	8	3518.27184	29
000	19	12	7	3518.27225	35
000	19	13	7	3704.23933	23
000	19	13	6	3704.23893	39
000	19	14	6	3898.99864	25
000	19	14	5	3898.99876	36
000	19	15	5	4101.34236	38
000	19	15	4	4101.34243	57
000	20	5	16	2771.28074	16
000	20	5	15	2878.89158	14
000	20	6	15	2895.35406	21
000	20	6	14	2958.14235	23
000	20	7	14	3012.50939	19
000	20	7	13	3034.08464	65
000	20	8	13	3134.84208	22
000	20	8	12	3138.72659	20
000	20	9	12	3270.57967	29
000	20	9	11	3271.01875	28
000	20	10	10	3421.33280	23
000	20	11	10	3585.59710	42
000	20	11	9	3585.59908	32
000	20	13	8	3948.27638	53
000	20	13	7	3948.27597	23
000	21	4	17	3011.92893	20
000	21	5	16	3134.35638	21
000	21	6	16	3145.24287	20
000	21	7	15	3267.12565	30
000	21	7	14	3299.50953	23
000	21	8	14	3390.67274	70
000	21	9	12	3526.82851	32
000	21	10	12	3676.04552	36
000	22	3	19	3107.86168	19
000	22	4	19	3107.93218	23
000	22	4	18	3263.76708	19
000	22	5	18	3264.62781	24
000	22	5	17	3397.63413	34
000	23	2	21	3176.56797	37
000	23	3	21	3176.56048	23

Table 2 (*continued*)

Vib	J	K_a	K_c	E^{emp}	dE
000	23	3	20	3359.25796	26
000	23	4	20	3359.29570	20
000	23	5	19	3524.81632	25
000	24	2	22	3428.11881	32
000	25	3	23	3688.74722	39
000	26	0	26	3485.83262	23
000	26	1	26	3485.83278	42
000	27	0	27	3746.40563	51
000	27	1	27	3746.40547	29
010	11	5	6	2243.40307	50
010	12	2	10	2201.54432	14
010	12	6	7	2493.95108	50
010	12	9	4	2904.45167	27
010	12	11	2	3244.86553	39
010	12	11	1	3244.86498	20
010	12	12	1	3430.59630	45
010	12	12	0	3430.59595	46
010	13	3	11	2355.28125	13
010	13	3	10	2438.02372	18
010	13	4	9	2496.84633	18
010	13	5	9	2550.38006	36
010	13	6	7	2656.40075	40
010	13	7	6	2775.97536	51
010	13	9	5	3065.06654	19
010	13	9	4	3065.06539	28
010	13	10	4	3229.97842	21
010	13	10	3	3229.97727	36
010	13	11	3	3406.40208	27
010	13	11	2	3406.40239	39
010	13	12	2	3592.82592	22
010	13	12	1	3592.82647	35
010	13	13	1	3787.93649	56
010	13	13	0	3787.93684	51
010	14	2	12	2513.36097	18
010	14	4	11	2622.71918	21
010	14	4	10	2679.51402	21
010	14	5	10	2722.87192	21
010	14	5	9	2741.77186	24
010	14	6	9	2828.43569	28
010	14	6	8	2831.37232	37
010	14	7	8	2948.82495	41
010	14	7	7	2949.06261	20
010	14	8	7	3085.74328	54
010	14	8	6	3085.75720	25
010	14	9	5	3237.73828	21
010	14	10	5	3402.99692	36
010	14	10	4	3402.99807	21
010	14	12	3	3767.03174	54
010	14	12	2	3767.03143	36
010	15	3	13	2683.38903	21
010	15	3	12	2793.69006	21
010	15	4	12	2800.62356	20
010	15	5	11	2906.61115	27
010	15	6	10	3013.91932	32
010	15	6	9	3019.81952	26
010	15	7	9	3133.96823	37
010	15	7	8	3134.59090	30
010	15	8	8	3270.47181	33
010	15	9	7	3422.40444	38
010	15	9	6	3422.40716	57
010	16	2	15	2716.88299	23
010	16	2	14	2861.17037	30
010	16	6	10	3222.09889	31
010	16	7	9	3332.67957	37
010	16	9	7	3619.01028	40
010	17	1	16	2894.03712	27
010	17	2	16	2894.04543	21
010	17	3	15	3048.97622	31
010	18	0	18	2896.14432	24
010	18	1	18	2896.14467	41
010	18	1	17	3080.50949	22
010	18	2	17	3080.51408	33
010	19	0	19	3081.26826	51
010	19	1	19	3081.26791	30

Notes: Vib – vibrational numbers, $J\ K_a\ K_c$ – rotational numbers, E^{emp} – empirical value of term value (in cm^{-1}), dE – uncertainty (2σ) of E^{emp} (in 10^{-5} cm^{-1} unit).

The comparison of the $D_2{}^{17}O$ empirical energies with previous sets reported by Toth [14], Tennyson et al. [1] and Kyuberis et al. [21] has been illustrated in Fig. 7 of Ref. [2]. Overall, the comparison with our new results is unchanged with the exception of two levels, (010) 12 $_{3\ 9}$ (E^{TW} = 2273.332 cm^{-1}) and (010) 13 $_{4\ 10}$ (E^{TW} = 2454.877 cm^{-1}), for which the differences compared to Ref. [21] are -0.05744 cm^{-1} and -0.05523 cm^{-1} instead of -0.11683 cm^{-1} and -0.11468 cm^{-1}, respectively.

We present in Fig. 2 a few examples of important deviations between the recorded spectra and the $D_2{}^{17}O$ line positions obtained by Kyuberis et al. [21]. The assignment of the problematic transitions is indicated. Kyuberis et al. provided a list of transition frequencies which mixes without distinction empirical, calculated using an effective Hamiltonian and variational transition frequencies. Nevertheless, the line list of Ref. [21] is given in HITRAN format [3]. So, the line parameters of each transition have uncertainty codes. Position uncertainty codes have values *ier* = 2, 3 and 4, the code *ier* = 4 corresponding to the smaller position uncertainty between 10^{-4} and 10^{-3} cm^{-1}. We assume that the *ier* = 4 code corresponds to the line position determined from the empirical values of energy. In the displayed examples of Fig. 2, the transition frequencies are given with a 10^{-3} cm^{-1} uncertainty (*ier* = 4) and believed to have an empirical origin, although the observed deviations largely exceed 10^{-3} cm^{-1}.

5. $D_2{}^{18}O$

5.1. Line positions

A detailed review of the previous reports of the FIR spectrum of $D_2{}^{18}O$ has been given in Ref. [2]. Only 128 pure rotational transitions were reported prior our analysis of SOLEIL spectra. These data were obtained by Johns [5] in the 51.5 – 216.4 cm^{-1} interval. This dataset was first extended using the analysis of a SOLEIL spectrum highly enriched in ^{18}O [22]: 138 line positions were reported between 51.538 and 294.093 cm^{-1} including 63 transitions absent in Johns' dataset. The set of measured $D_2{}^{18}O$ line positions was then considerably enlarged from 191 to 1175 transitions using the SOLEIL analysis of the "D_2O" spectra #15-18 [2] and the "HDO" spectra #19-21 [10]. To the best of our knowledge, there are no other measurements of $D_2{}^{18}O$ absorption spectrum in the range of 50 – 720 cm^{-1}.

In Ref. [2], the maximum $D_2{}^{18}O$ partial pressure was about 33.6 μbar (corresponding to 0.42% at 8 mbar in spectrum #17) while in the presently analyzed spectrum #14, the $D_2{}^{18}O$ partial pressure is about 273 μbar (corresponding to 3.4% at 8 mbar – see Table 1). As a result, the detectivity threshold is lowered compared to previous studies and 1523 $D_2{}^{18}O$ transitions are measured (to be compared to 1169 in Ref. [2] and 704 in Ref. [10]). A total of 375 transitions are new compared to Refs. [2,10,22]. They include 113 rotational transitions of the ν_2-ν_2 band (236 and 81 transitions of the ν_2-ν_2 band were reported in Ref. [2] and in Ref. [10], respectively) which leads to a total of 353 measured line positions for the ν_2-ν_2 band. Eight transitions of the $2\nu_2$-$2\nu_2$ rotational band are observed for the first time. Note that, due to accidental overlapping with stronger lines, 24 $D_2{}^{18}O$ transitions reported in Ref. [2] could not be measured in the present spectra. The overview of the SOLEIL $D_2{}^{18}O$ dataset including a total of 1549 transitions between 50.58 and 666.03 cm^{-1} is displayed on Fig. 3.

The comparisons of the $D_2{}^{18}O$ measured line positions to their determinations in Ref. [2] ($\nu^{\#15\text{-}18}$) and Ref. [10] ($\nu^{\#19\text{-}21}$) give a very satisfactory agreement (see supplementary material **SM1**). The corresponding $RMS = \sqrt{\sum_{i=1}^{N}\left(\nu_i^{TW} - \nu_i^{\#15-18}\right)^2/N}$ value is 16.0 × 10^{-5} cm^{-1} for N = 1141. The largest discrepancy $d\nu = |\nu^{TW} - \nu^{\#15\text{-}18}|$ of -175 × 10^{-5} cm^{-1} is due to an erroneous assignment of the (010) 10 $_{7\ 4}$ – (010) 9 $_{6\ 3}$ transition in Ref. [2]. In this reference, this transition was given as second assignment of a line at 243.57258 cm^{-1} (the first assignment being the 5 $_{5\ 1}$ – 4 $_{2\ 2}$ transition of $D_2{}^{17}O$) while a line at 243.57089 cm^{-1} was left unassigned. The latter is indeed due to the (010) 10 $_{7\ 4}$ – (010) 9 $_{6\ 3}$ transition of $D_2{}^{18}O$ in agreement with the present observation at 243.57082 cm^{-1}.

The two large $d\nu$ values of 0.02347 cm^{-1} for the (010) 13 $_{12\ 2}$ – (010) 12 $_{11\ 1}$ and (010) 13 $_{12\ 1}$ – (010) 12 $_{11\ 2}$ rotational doublet are not included in the statistics above. These differences are due to an erroneous assignment of the doublet to a weak line at 344.78259 cm^{-1} in spectrum #17 of Ref. [2]. This

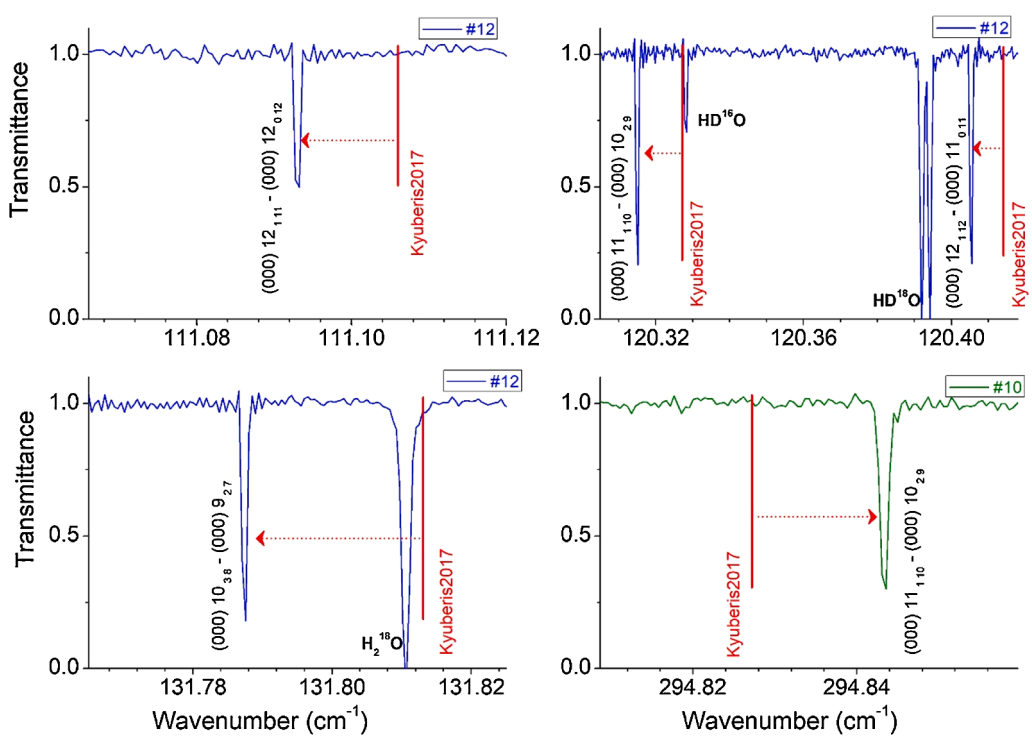

Fig. 2. Examples of important deviations between the recorded spectra and the $D_2{}^{17}O$ line positions obtained by Kyuberis et al [21]. The assignment of the problematic transitions is indicated.

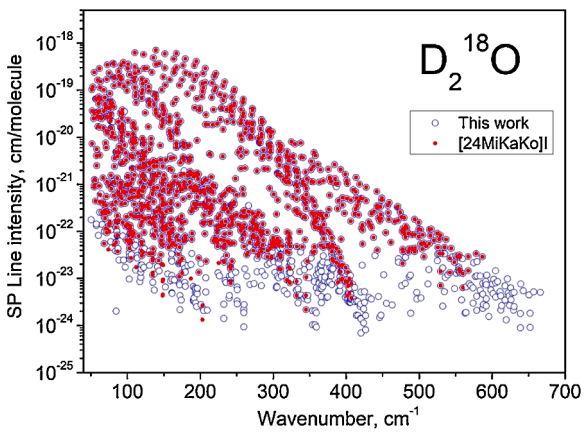

Fig. 3. Overview of the $D_2{}^{18}O$ transitions measured in this work (spectra #10–14, blue circles) and in Ref [2] ([24MiKaKo]I) (spectra #15–18, red dots). The measured line positions have been associated with the SP intensities (for pure $D_2{}^{18}O$).

"line" is believed to be an experimental artifact as it is not reproduced in any other SOLEIL spectrum. The actual position of the considered doublet is presently determined at 344.80607 cm^{-1} from spectra #13 and #14.

The comparison with Ref. [10] leads to an $RMS = \sqrt{\sum_{i=1}^{N}\left(\nu_i^{TW} - \nu_i^{\#19-21}\right)^2/N}$ value of 13.8×10^{-5} cm^{-1} for $N = 703$. The largest discrepancy $d\nu = |\nu^{TW} - \nu^{\#19\text{-}21}|$ of 94×10^{-5} cm^{-1} is due to the 15 $_{10\ 5}$ – 14 $_{9\ 6}$ transition at 333.31149 cm^{-1} which was assigned as a component of an unresolved doublet (15 $_{10\ 5}$ – 14 $_{9\ 6}$ and 15 $_{10\ 6}$ – 14 $_{9\ 5}$) in spectrum #19 of Ref. [10].

5.2. Energy levels

As for $D_2{}^{17}O$, the RITZ computer code was used to retrieve the empirical values of the energy levels of the ground and (010) states of $D_2{}^{18}O$. The set of used transition frequencies differs significantly from that used in Ref. [2]. First of all, we considered only the transitions between the five lower vibrational states: (000), (010), (020), (100), and (001), thus excluding more than 5500 near-infrared transitions from Refs. [15,16,23,24] between 5972 and 9109 cm^{-1} which were used in Ref. [2]. On the other hand, 138 line positions of Ref. [22] were taken into account leading to a global set of more than 9700 transitions between 0.29 and 3218 cm^{-1}. The dataset contains 21 microwave transitions given by Bellet et al. [12]; 10 and 144 FIR transitions by Fleming & Gibson [25] and Johns [5], respectively; more than 2300 ν_2 transitions from Refs. [26–29]; about 3800 transitions of the $2\nu_2$, ν_1 and ν_3 bands by Toth [30] and Ni et al. [31] and more than 3500 rotational transitions from SOLEIL spectra of Ref. [22] (^{18}O enriched spectrum), Ref. [2] (spectra #15-18), Ref. [10] (spectra #19-21) and this work (spectra #10-14).

The term values of a total of 1751 levels of the first five vibrational states of $D_2{}^{18}O$ were determined. They include 452 term values of the ground vibrational state up to $J_{max} = 27$, $K_{a\ max} = 17$ and $E_{max} = 4023$ cm^{-1} and 302 term values of the first excited vibrational state up to $J_{max} = 22$, $K_{a\ max} = 13$ and $E_{max} = 3906$ cm^{-1}. The complete set of energy levels of the ground and first excited vibrational states is provided as a separate supplementary material (**SM3**). Compared to the set derived in Ref. [2], 53 levels are new and two are corrected for the ground state. Six levels are new and 15 are corrected for the (010) state. Newly determined term values are listed in Table 3. Note that the 23 $_{5\ 19}$ – 22 $_{4\ 18}$ pure rotational transition at 260.02455 cm^{-1} is not included in the main spectroscopic network. So, the energies of the 22 $_{4\ 18}$ and 23 $_{5\ 19}$ ground state levels could not be derived.

The levels corrected compared to Ref. [2] correspond to $dE = |E^{TW} - E^{Ref.[2]}|$ differences both larger than 30×10^{-5} cm^{-1} and larger than twice their uncertainty given in Ref. [2]. (Note that in the present work, the level uncertainty is defined as the 2σ standard deviation of the energy determination, while in Ref. [2], the 1σ standard deviation was chosen). The two corrected ground state energies are those of the 19 $_{6\ 14}$ and 20 $_{5\ 15}$ levels. Their corresponding energy differences (in 10^{-5} cm^{-1} units), are 121 and 96 with uncertainties of 16 and 20 respectively. Among the fifteen corrected levels of the first excited vibrational state, the dE deviations of ten levels (13 $_{11\ 3}$, 14 $_{8\ 7}$, 14 $_{8\ 6}$,

Table 3
Empirical energies of newly determined rotational levels of $D_2{}^{18}O$ in the ground and first excited vibrational states.

Vib	J	K_a	K_c	E^{emp}	dE
000	17	16	2	3815.06680	30
000	17	16	1	3815.06685	48
000	17	17	1	4022.79342	32
000	17	17	0	4022.79354	50
000	18	9	10	2781.38006	42
000	18	14	5	3637.06984	49
000	18	14	4	3637.06975	35
000	18	15	4	3835.04042	56
000	18	15	3	3835.04039	42
000	19	6	13	2697.70105	16
000	19	9	10	3013.45588	79
000	19	10	10	3161.64754	19
000	19	10	9	3161.66576	38
000	19	14	6	3870.33622	38
000	19	14	5	3870.33610	55
000	20	4	17	2624.02839	14
000	20	6	15	2886.89549	12
000	20	6	14	2952.83718	24
000	20	7	14	3003.15495	20
000	20	8	13	3123.57944	18
000	20	8	12	3128.09173	19
000	20	9	12	3256.78395	23
000	20	9	11	3257.31274	20
000	20	10	11	3404.72945	84
000	20	10	10	3404.77318	19
000	20	11	10	3566.12845	50
000	20	11	9	3566.13035	27
000	20	12	9	3739.27002	51
000	20	12	8	3739.26820	33
000	20	13	8	3922.63489	74
000	20	13	7	3922.63392	52
000	21	3	18	2856.36173	22
000	21	4	17	3003.04588	18
000	21	5	16	3126.69762	18
000	21	6	16	3136.11764	19
000	21	6	15	3218.27204	20
000	21	7	15	3257.37537	24
000	21	7	14	3292.85596	23
000	21	10	12	3659.31876	28
000	22	2	20	2924.36076	18
000	22	3	20	2924.36096	27
000	22	4	19	3097.96234	22
000	22	5	18	3254.66297	22
000	22	5	17	3388.79218	18
000	22	6	16	3492.85450	19
000	23	2	21	3165.98929	35
000	23	3	20	3348.44263	25
000	23	4	20	3348.47357	23
000	24	3	21	3607.94356	33
000	26	0	26	3473.94419	30
000	26	1	26	3473.94427	58
000	27	0	27	3733.64051	62
000	27	1	27	3733.64043	35
010	13	13	1	3758.82127	29
010	13	13	0	3758.82166	34
010	14	12	3	3740.84724	34
010	14	12	2	3740.84727	25
010	15	11	5	3741.91260	35
010	15	11	4	3741.91286	20

Notes: Vib – vibrational numbers, $J\ K_a\ K_c$ – rotational numbers, E^{emp} – empirical value of term value (in cm^{-1}), dE – uncertainty (2σ) of E^{emp} (in 10^{-5} cm^{-1} unit).

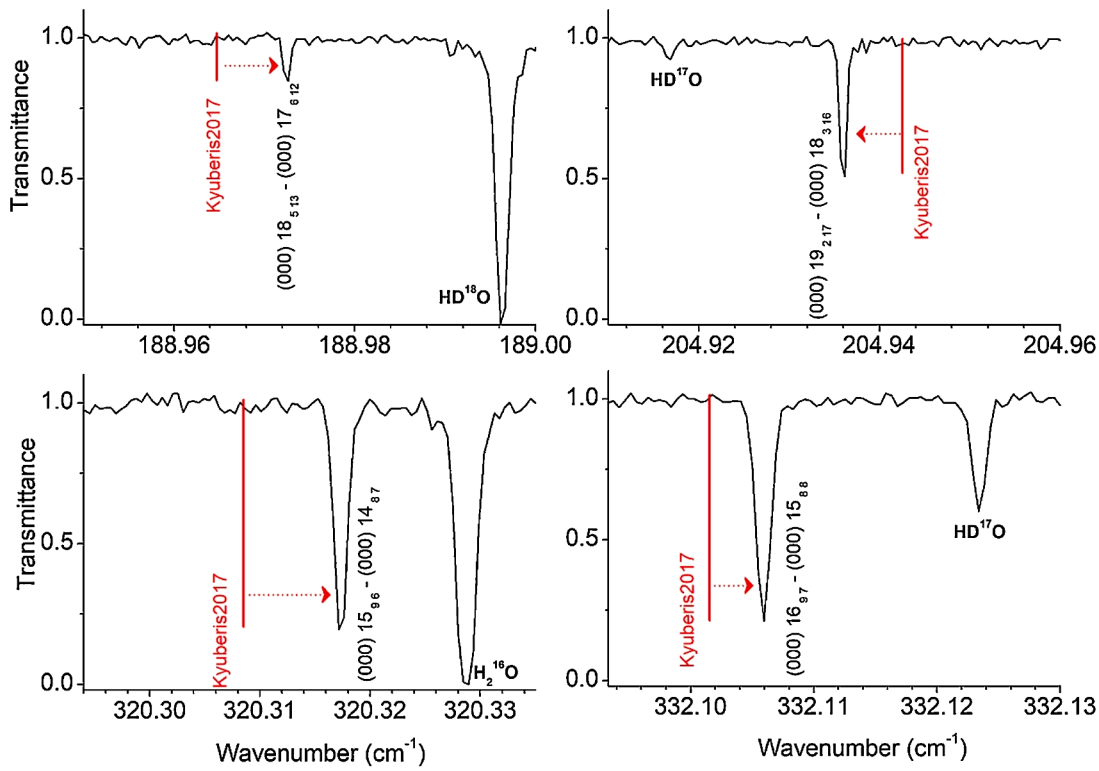

Fig. 4. Examples of important deviations between the recorded spectra and the $D_2{}^{18}O$ line positions obtained by Kyuberis et al [21] (Kyuberis2017). The assignment of the problematic transitions is indicated.

$14_{9\,5}$, $15_{0\,15}$, $15_{8\,8}$, $16_{1\,16}$, $17_{2\,16}$, $17_{4\,14}$, and $19_{2\,18}$) range between 33 and 67 (in 10^{-5} cm^{-1} units). All these differences are due to the fact that, compared to Ref. [2], the currently used set of transition frequencies include much more high-precision data from SOLEIL spectra. For the (010) $13_{12\,2}$ and $13_{12\,1}$ levels, the *dE* differences exceed 0.023 cm^{-1} due to an erroneous assignment of the (010) $13_{12\,2}$ – (010) $12_{11\,1}$ and (010) $13_{12\,1}$ – (010) $12_{11\,2}$ doublet discussed above. The energy differences of the three remaining corrected levels of the (010) state ($19_{1\,18}$, $20_{0\,20}$, $20_{1\,20}$ with $dE = 0.00289$, 0.00752 and 0.00762 cm^{-1}, respectively) are due to the fact that in Ref. [2], the corresponding energies were determined from the frequencies of unresolved doublets of the ν_2 band (from Refs. [26,27]), but the SOLEIL spectra provide now accurate frequencies of well resolved ν_2 - ν_2 rotational transitions, allowing for a more accurate determination of the energy of these levels.

Excluding the corrected levels, the comparison to the empirical values of Ref. [2] included in the **SM3** supplementary material shows a satisfactory agreement. An *RMS* value of 12.8×10^{-5} cm^{-1} is achieved for the differences $dE = E^{TW} - E^{Ref.[2]}$ (N= 690).

As for $D_2{}^{17}O$, the comparison of the (000) and (010) empirical energies of $D_2{}^{18}O$ with previous sets reported in by an IUPAC task group [1], Liu et al. [29] and Kyuberis et al. [21] has been included in Ref. [2] (see Fig. 8 of this reference). Overall, the comparison with our new results is unchanged. We present in Fig. 4 a few examples of important deviations between the recorded spectra and the $D_2{}^{18}O$ line positions obtained by Kyuberis et al. [21]. The assignment of the problematic transitions is indicated. Similarly to Fig. 2, the uncertainty code (*ier* = 4) of the displayed line positions of Ref. [21] corresponds to position uncertainties of 10^{-3} cm^{-1}, which is largely smaller than the observed deviations to the measured values.

6. Recommended line lists

The obtained term values of the ground and first excited states of $D_2{}^{17}O$ and $D_2{}^{18}O$ were used for to generate recommended line lists of the (000) – (000) and (010) – (010) rotational and of the ν_2 fundamental bands of these molecules. The variation lists calculated by S. Tashkun based on the results of Partridge and Schwenke [7,8] were used as basis. The general procedure of the line list construction is described in Ref. [2]. It consists in replacing the SP variational position values by their empirical counterpart calculated by difference of empirical term values when available. The main lists (**SM4** and **SM5** for $D_2{}^{17}O$ and $D_2{}^{18}O$, respectively) cover the 0 – 1850 cm^{-1} range and corresponds to an intensity cut-off of 1×10^{-26} cm/molecule at 296 K (for the pure species). The transitions remaining with variational positions are weak or very weak (see Fig. 6 of Ref. [2]). Due to the larger number of known empirical energy levels, compared to Ref. [2], the recommended lists contain fewer transitions with (less accurate) variational frequencies. For $D_2{}^{17}O$, the number of variational positions is reduced from 2785 to 1705 (over a total of 7046 transitions). For $D_2{}^{18}O$, the number of variational positions is reduced from 1463 to 1199 (over a total of 7084 transitions).

Two additional lists (**SM6** and **SM7** for $D_2{}^{17}O$ and $D_2{}^{18}O$, respectively) are generated and provided as supplementary materials for the microwave transitions (between 2.7 GHz and 1.5 THz) with an intensity cut-off lowered to 1×10^{-29} cm/molecule.

7. Conclusion

From the analysis of five FIR spectra (#10-14) recorded at SOLEIL synchrotron with a high enrichment in ^{17}O, ^{18}O and deuterium, it has been possible to significantly improve our knowledge of the rotational levels of the ground and first excited states of $D_2{}^{17}O$ and $D_2{}^{18}O$. This

study adds 548 and 380 newly measured transitions for $D_2{}^{17}O$ and $D_2{}^{18}O$, respectively, leading to the new determination of 147 and 59 energy levels, respectively. Merging these results with literature sources, in particular Ref. [2] from the analysis of SOLEIL spectra #15-18, the sets of empirical rotational-vibrational energies for the ground, (000), and first excited, (010), vibrational states have been updated and recommended lists of transitions have been generated for the (000) – (000), and (010) – (010) rotational bands and the ν_2 fundamental band of $D_2{}^{17}O$ and $D_2{}^{18}O$.

The present contribution is the last one dedicated to the analysis of 21 high sensitivity spectra (#1-21) of water vapor samples recorded at SOLEIL synchrotron with various isotopic contributions: natural abundance (#1-5) [32], ^{17}O enriched (#6-9) [33], D_2O (#15-18) [2], "HDO" obtained with a 1:1 mixture of H_2O and D_2O (#19-21) [10] and strong enrichment in both ^{17}O, ^{18}O and deuterium [6] and present work. As a result of the brightness of the synchrotron radiation and of the long absorption pathlength (151.75 m), the sensitivity of the recorded SOLEIL spectra improved by about three orders of magnitude compared to that of all previous absorption studies available in the region. For all the nine water vapor isotopologues, large sets of newly measured line positions were derived with a typical accuracy of 5×10^{-5} cm^{-1} for isolated lines. A significant number of new energy levels were determined and a general improvement of the accuracy of the energy levels of the ground and first excited vibrational states were achieved for $H_2{}^{16}O$ [32], $H_2{}^{17}O$ and $HD^{17}O$ [33], $H_2{}^{18}O$ [22], $D_2{}^{X}O$ (X = 16-18) ([2] and this work), $HD^{16}O$ [10], $HD^{17}O$ and $HD^{18}O$ [6].

It is worth to underline that the lower and upper states of the observed FIR transitions are lower states of transitions observed in all the spectral ranges. Thus, the determination or correction of energy levels based on the studied FIR spectra propagates over the whole frequency range of the water vapor spectrum. This gives a particular importance to the FIR region in the elaboration of spectroscopic databases.

The publication of these results in the Memorial Issue of JQSRT in Honor of Mikhail Tretyakov is particularly justified. Indeed, our project dedicated to the FIR rovibrational transitions of water vapor studied at SOLEIL follows the project headed by Prof. Mikhail Tretyakov dedicated to the study of the weak absorption continuum of water vapor using the SOLEIL synchrotron source [34–38]. During the recordings of the SOLEIL spectra dedicated to the self-continuum of $H_2{}^{18}O$ [34], significant deviations were noted for the positions of the $H_2{}^{18}O$ rovibrational lines. This observation led to a first publication on the FIR rovibrational transitions of $H_2{}^{18}O$ [22] which was a side-product of the continuum study and motivated our application a one-week measurement campaign dedicated to the RV transitions of the nine stable water isotopologues (SOLEIL Project No20210051). We are thus indebted to Prof. Mikhail Tretyakov to have involved us in the study of the FIR spectrum of water vapor using the synchrotron radiation source.

CRediT authorship contribution statement

S.N. Mikhailenko: Writing – original draft, Investigation, Data curation, Conceptualization. **A.O. Koroleva:** Investigation. **A. Campargue:** Writing – review & editing, Writing – original draft, Supervision, Investigation, Funding acquisition.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Supplementary materials

Supplementary material associated with this article can be found, in the online version, at doi:10.1016/j.jqsrt.2025.109638.

Data availability

The data that support the findings of this study are available within the article and its supplementary materials.

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